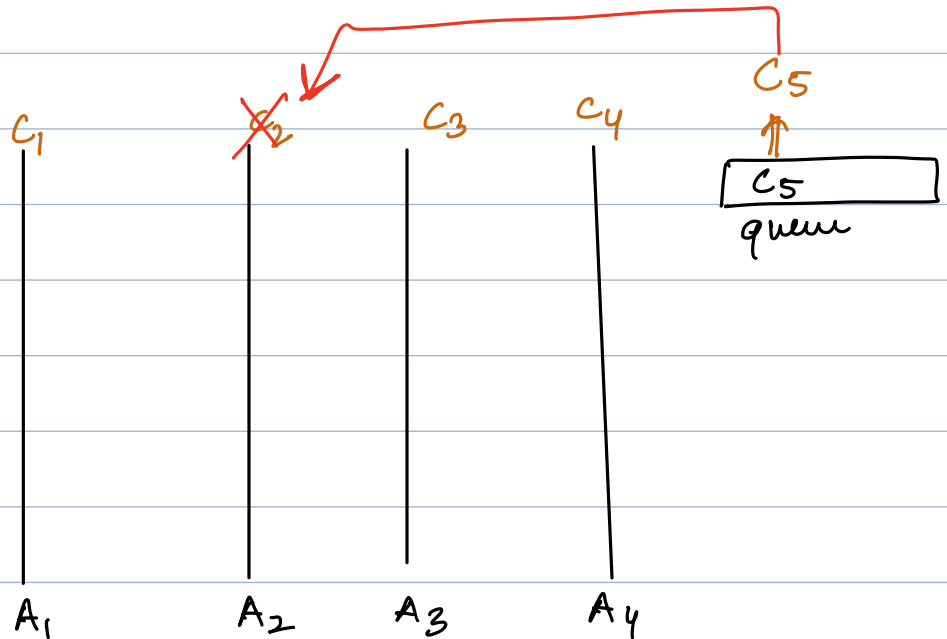


Recap

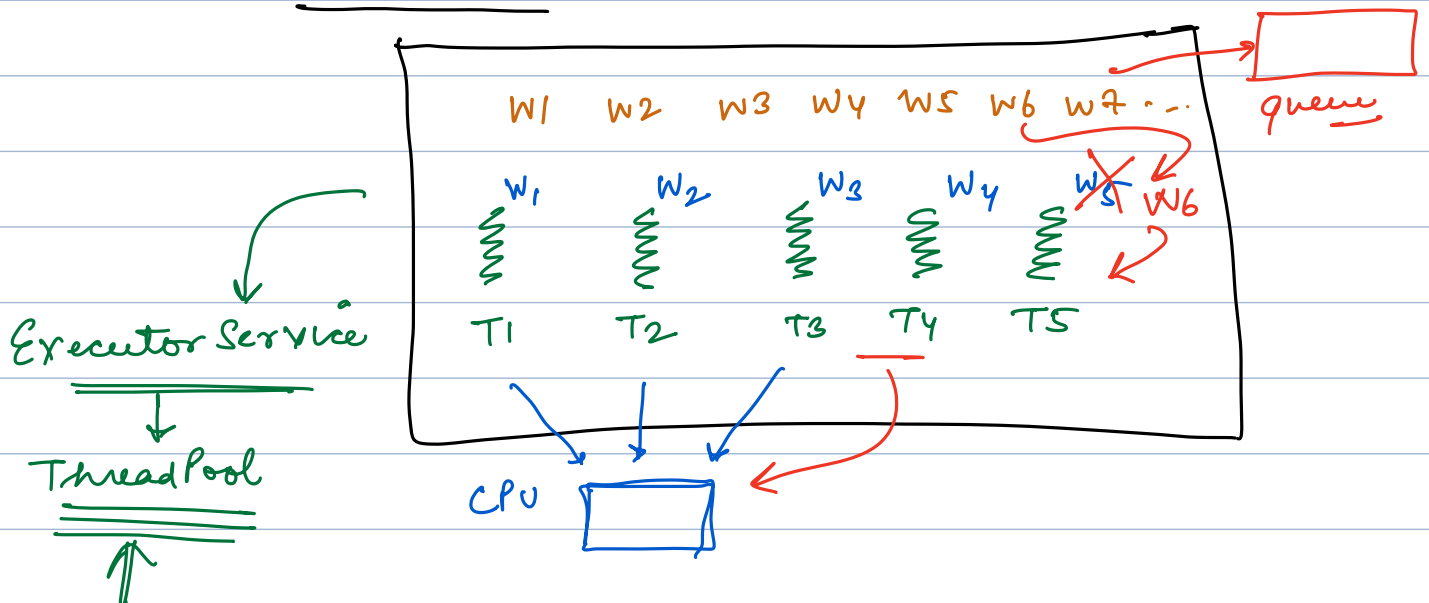
Runnable
extending Threads
Print 1 → 100

→ { Executors
callable.
Merge-Sort.

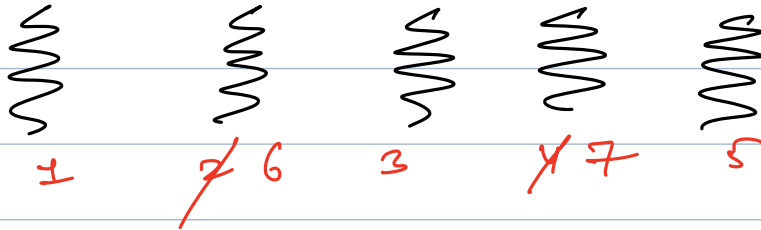
1 → 1,000,000 X



ThreadPool



new Fixed Thread Pool (5)



for ($i = 1 \rightarrow \underline{10^6}$)

~~Thread t = new Thread();~~

Fixed

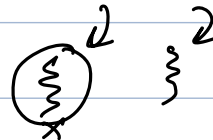


no of Threads
will always
be fixed

cached



If all threads are busy,
it will create a
new thread.



Runnable

↑
X

void run()

callable interface

generic

T call() {

}

If you want to
return some
data from
Thread

MergeSort

{

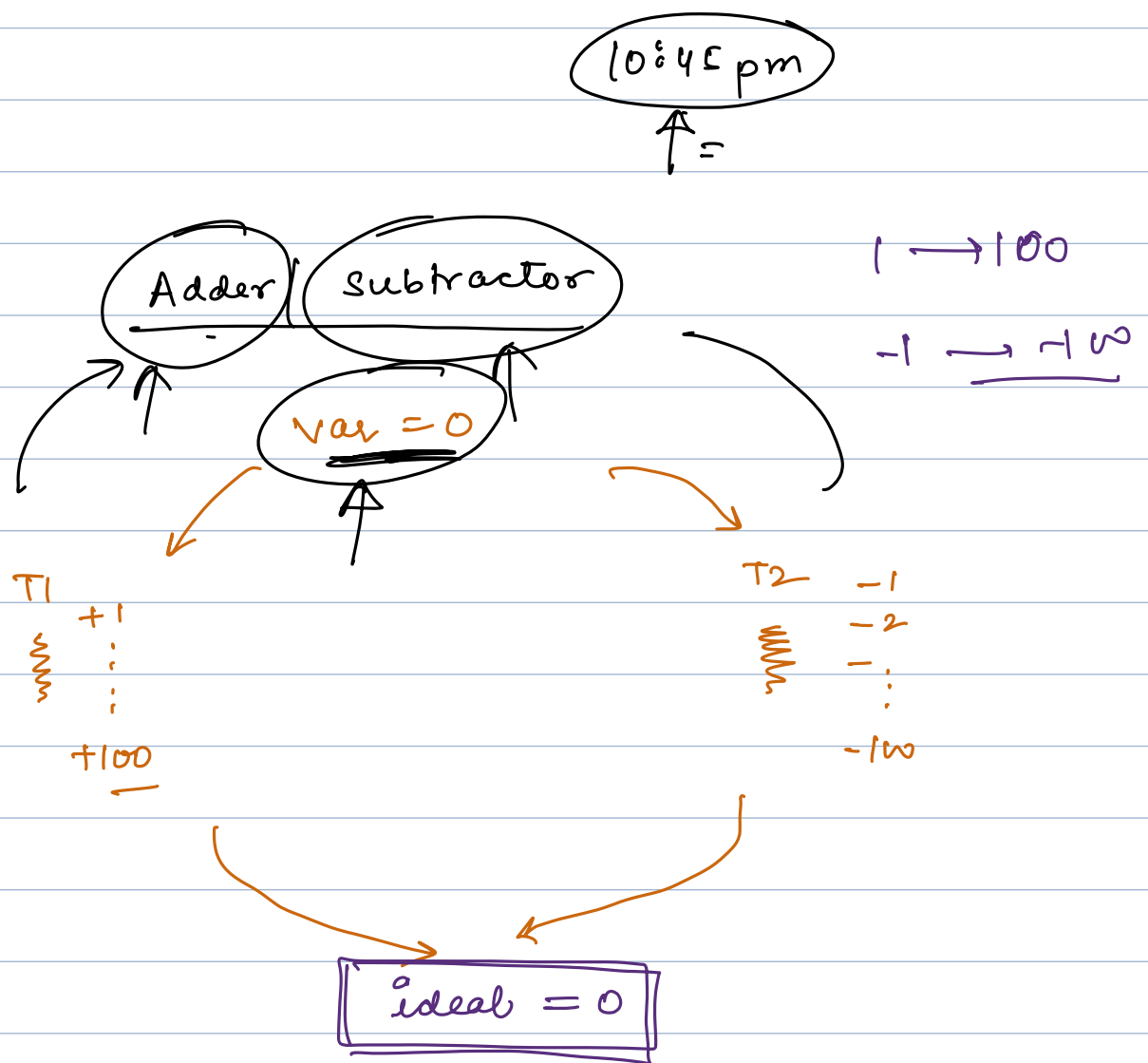
1 2 3 5 6 7 8 9
8 1 2 9 6 3 7 5

~~1~~ ~~2~~ 8 9
8 1 2 9

~~3~~ ~~5~~ ~~6~~ ~~7~~
6 3 7 5

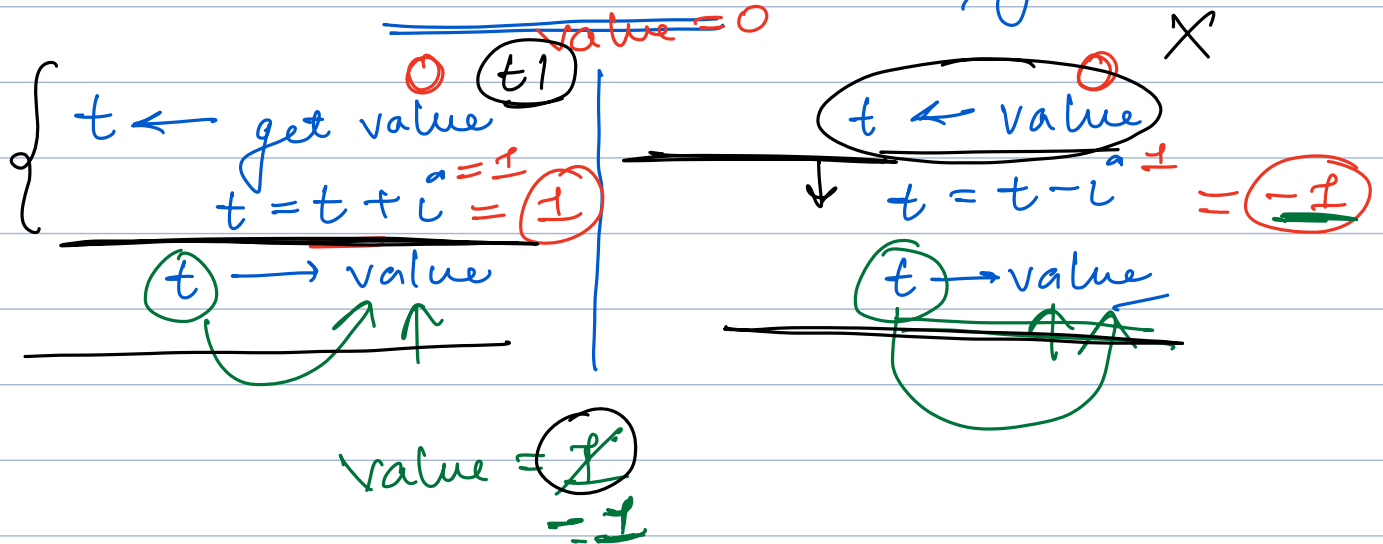
~~1~~ ~~8~~
8 1
8 1

~~2~~ ~~9~~
2 9
2 9



$$\begin{array}{rcl}
 \left\{ \begin{array}{cccccc} 1 & -1 & -2 & -3 & +2 & +3 \\ 1 & +2 & +3 & -1 & -2 & -3 \end{array} \right. & = & \begin{array}{c} 0 \\ 0 \end{array}
 \end{array}$$

$value += i$ — single



$-1 - 100$
 $\quad \quad \quad 50$
 100×101
 $\quad \quad \quad 21$
 ~ 2000

Sync issue

- ⇒ {
- critical section
 - Race condⁿ
 - Preemptiveness